A Theory of Economic Slack

Pascal Michaillat

Draft version: July 2026
Draft URL: pascalmichaillat.org/18/

4 SLACKISH MARKET MODEL 3
4.1 Buyers and sellers 3
4.2 Matching function 4
4.3 Market tightness 4
4.4 Rate of slack 5
4.5 Market supply 7
4.6 Matching wedge 8
4.7 Market demand 10
4.8 Market price 15
4.9 Solution of the model 16
4.10 Model with fixed prices 21
4.11 Summary 25

BIBLIOGRAPHY 27

CHAPTER 4.

Slackish market model

This chapter presents the most basic model of a slackish market. In a slackish market, the market tightness, not the market price, adjusts to equilibrate supply and demand and to determine allocations and welfare. The market price is governed by a price norm, rather than by a market-clearing condition.

The model presented in this chapter is basic in many ways, and it will be extended in the next chapters. For example, this chapter's model is static—it lasts only one period. Of course, the real world is dynamic, so in chapter 8 we provide a dynamic extension of the model and show how we can move from a static to a dynamic slackish model. The dynamic extension allows us in particular to introduce long-term relationships.

4.1. Buyers and sellers

We assume that there are many sellers, each indexed by $i \in [0,1]$. Each seller sells a fixed amount of goods $k_i > 0$. The aggregate amount of goods for sale on the market is $k = \int_0^1 k_i \, di$. The quantity $k > 0$ represents the market capacity.

We assume that there are many buyers as well, each indexed by $j \in [0,1]$. Since the market is slackish and not Walrasian, it is difficult for buyers to find the goods they like. Thus, buyers need to visit various sellers to buy goods. The number of sellers that buyer j visits is denoted by v_j. The aggregate number of visits to sellers on the market is $v = \int_0^1 v_j \, dj$.

4.2. Matching function

First, we examine how sellers are able to actually find buyers for their goods. Based on the number of goods that are available, k, and the number of visits, v, we can determine the number of trades that are realized. Here, a trade is one good being bought at price $p > 0$. The number of trades, which is the output and sales in the model, is denoted by y.

Because we are working with a slackish market, the number of trades is determined by a matching function m. In this market, the demand-side input into the matching function is the aggregate number of visits to buy goods, v, rather than the mass of buyers (normalized to 1); the supply-side input is the aggregate number of goods for sale, k, rather than the mass of sellers (also normalized to 1). Hence, the number of trades is given by

$$y = m(k, v). \tag{4.1}$$

We assume that the matching function satisfies the generic properties listed in chapter 3. In particular, since we are in a static model, the number of trades cannot be more than the minimum of the number of visits and the number of goods available, so the matching function satisfies $y \leq \min(k, v)$.

4.3. Market tightness

As we saw in chapter 3, the trading probabilities are solely determined by the market tightness, θ, which here is the aggregate number of visits divided by the aggregate number of goods for sale:

$$\theta = \frac{v}{k}. \tag{4.2}$$

The selling probability is given by $f(\theta) = m(1, \theta)$. It satisfies $f(0) = 0$ and is increasing and concave in θ. These properties say that it is impossible to sell without visits and that it is easier to sell in a tighter market—when visits are in larger numbers relative to goods for sale.

The buying probability, which is the probability that a visit to a seller is successful, is given by $q(\theta) = m(1/\theta, 1)$. It is decreasing in θ, and $q(\theta) \to 0$ when $\theta \to \infty$. These properties say that a visit is less likely to be successful in a tighter market—as there is more competition for available goods—and that it becomes impossible to buy without goods for sale.

This is a general property of slackish markets: when the market is tighter, it is a better time to be a seller: goods are more likely to be sold. On the other hand, when the market is slacker, it is a good time to be a buyer: visits are more likely to be successful. Figures 4.1A and 4.1B display the selling and buying probabilities for an illustrative calibration.

4.4. Rate of slack

Because we are working with a slackish market, some goods always remain unsold. We define the rate of slack as the share of goods that remain unsold. It is easy to compute this slack rate and express it as a function of market tightness.

We previously saw that each good is sold with probability $f(\theta)$. For a seller with k_i goods for sale, omitting randomness at the seller level, the number of goods sold is $y_i = f(\theta)k_i$. The aggregate number of sales on the market is $y = \int_0^1 y_i \, di$, which can be expressed as

$$y = \int_0^1 f(\theta)k_i \, di = f(\theta) \int_0^1 k_i \, di = f(\theta)k. \tag{4.3}$$

Here we simply recover the matching function from the individual selling probabilities, since $f(\theta)k = m(1, \theta)k = m(k, \theta k) = m(k, v)$.

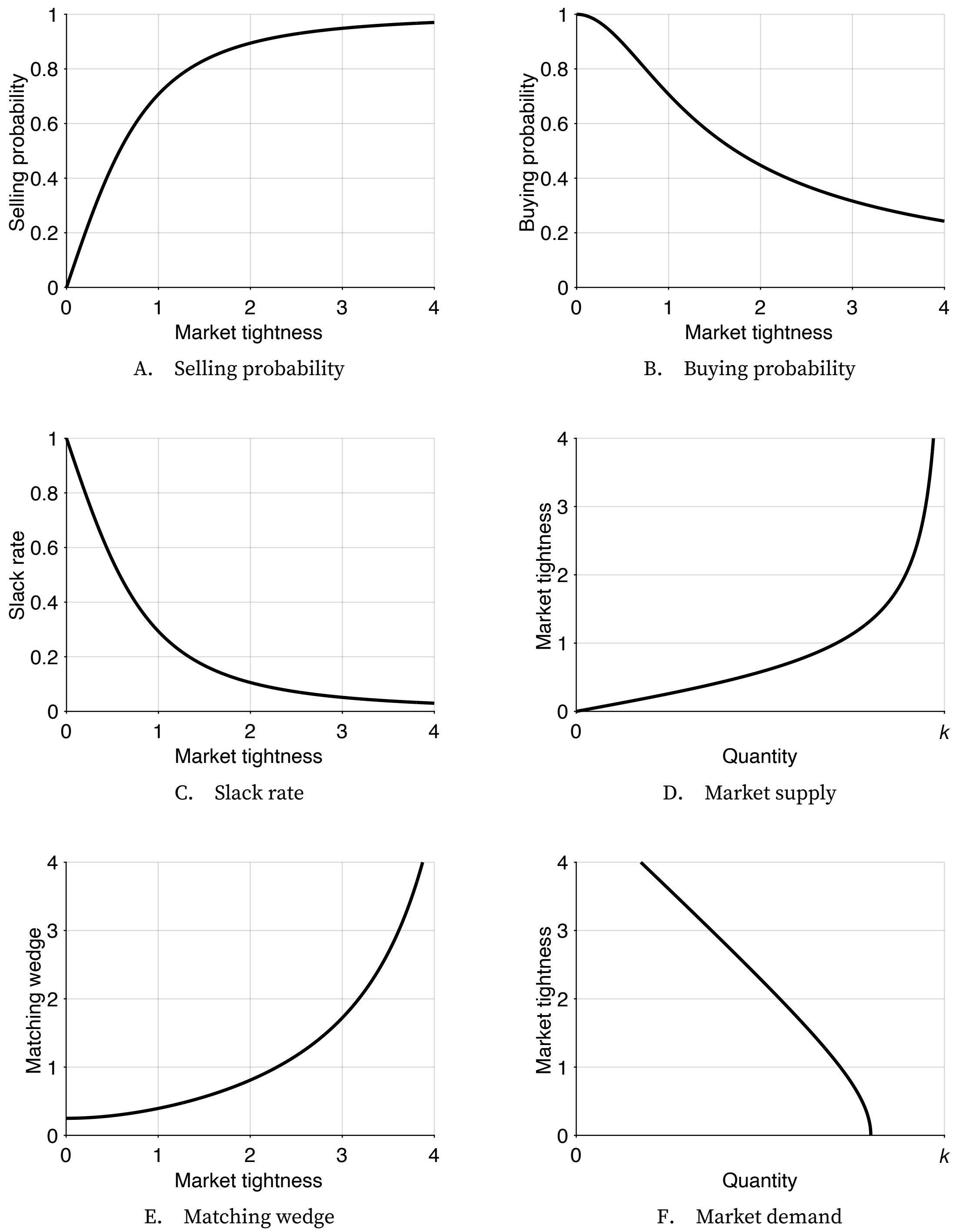
We define the rate of slack u in the market as the share of goods that remain unsold. In the aggregate, a share $f(\theta) \in [0, 1]$ of goods is sold, so the slack rate is a simple function of market tightness:

$$u(\theta) = 1 - f(\theta). \tag{4.4}$$

The properties of the slack rate follow from those of the selling probability $f(\theta)$. The slack rate is 1 when tightness is 0, because the selling probability is 0 when tightness is 0: when there are no buyers, nothing gets sold. The slack rate then decreases with tightness, because the selling probability increases with tightness. And since the selling probability is concave in tightness, the slack rate is convex in tightness.

The rate of slack can be interpreted in different ways depending on the type of goods sold on the market. If labor is sold on the market, then the rate of slack is just the rate of unemployment. If services are sold, then the rate of slack indicates the share of time that sellers are idle. If durable goods are sold, then the rate of slack gives the share of goods that are added to the unsold inventory. If perishable goods are sold, then the rate of slack indicates the share of goods that go to waste.

Figure 4.1C displays the rate of slack as a function of tightness to illustrate its properties. The figure also shows that the basic slackish model produces a Beveridge curve: a negative relationship between slack rate and visit rate. Indeed, in this static model, the visit rate is the number of visits per goods for sale, v/k, so the visit rate coincides with the market tightness. Therefore, the negative relationship between slack rate and market tightness corresponds to a Beveridge curve. In this static model the logic behind the Beveridge curve is trivial: a higher visit rate means a higher market tightness, which implies a higher selling probability, and thus fewer unsold goods and a lower slack rate.



A. Selling probability

B. Buying probability

C. Slack rate

D. Market supply

E. Matching wedge

F. Market demand

FIGURE 4.1. Numerical illustration of the slackish market model

The selling probability is given by (3.4). The buying probability is given by (3.5). The slack rate is given by (4.4). The market supply is given by (4.5). The matching wedge is given by (4.8). The market demand is given by (4.15). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, $a = 2$, and $p = 1$.

4.5. Market supply

We now compute the market supply, which is the first key relation in analyzing how a slackish market behaves.

4.5.1. Expression of the market supply

A key distinction in this model is between the notional supply of goods and the effective supply. On the market, we have k goods that sellers would like to sell, which is the notional market supply. Here the notional supply is fixed, but it could depend on the market price and tightness, as we will see in chapter 7, where sellers determine the amount of goods that they bring on the market based on expected market conditions.

However, in the model, we use the effective market supply, which is the amount of goods actually sold given tightness, and which is less than the notional supply. The gap arises because the amount of goods that sellers are actually able to sell is less than the amount of goods that they put on the market, since, in our slackish market, each good is only sold with a probability that is less than one.[1]

By combining the sales equation (4.3) and slack equation (4.4), we obtain a simple expression for the market supply:

$$y^s(\theta) = [1 - u(\theta)]\, k, \tag{4.5}$$

where $u(\theta)$ is the slack rate and k is the market capacity. The market supply does not depend on the market price at all. To formalize the definition of market supply: it is just the amount of goods sold given the matching structure and the amount of goods supplied to the market by sellers. We have a gap between notional supply and effective supply that we use in our model, because not all goods supplied are traded: $u(\theta) > 0$ so $y^s(\theta) < k$.

4.5.2. Properties of the market supply

Let us now examine the properties of the market supply, given by (4.5). It is clear that the properties of the market supply follow from the properties of the slack rate, $u(\theta)$. When market tightness is 0, the slack rate is 1, so the supply is 0. Then, the supply is increasing in tightness, because the slack rate is decreasing in tightness. Finally, the supply is concave in tightness because the slack rate is convex in tightness.

We plot the market supply curve in a tightness-quantity plane to visualize its behavior (figure 4.1D). In Walrasian models, supply and demand curves are represented in price-

[1] The terminology of notional and effective supply and demand is borrowed from the Old Keynesian model, although the terms were used differently in that model: the notional supply and demand were Walrasian supply and demand, while the effective supply and demand took into account rationing in the other markets (Barro and Grossman 1971).

quantity planes. But in slackish models, tightness is the main equilibrating variable, so we conduct most of the analyses in tightness-quantity planes.

Because the function $y^s(\theta)$ is concave in θ, but the axes are inverted in the tightness-quantity plane (in the sense that tightness is on the y-axis instead of the x-axis, and quantities on the x-axis instead of the y-axis), then the supply curve appears convex.

4.6. Matching wedge

We have seen that buyers visit sellers to trade with them. These visits are required so buyers and sellers might match. Visiting sellers is costly: on each visit, a buyer incurs a matching cost $\kappa \in (0, q(0))$. The matching cost κ represents the number of goods spent on each visit. In the best-case scenario, a visit is successful with probability $q(0)$, so in expectation a visit yields $q(0)$ goods. If the matching cost is more than $q(0)$ to begin with, no buyer would ever try to buy anything. Thus, we impose $\kappa < q(0)$.

4.6.1. Expression of the matching wedge

The matching cost creates a wedge between the quantity of goods a buyer purchases and the quantity they actually consume. Indeed, when a buyer purchases goods, they use them for two purposes. First, they consume some of the goods, which provide utility. We denote buyer j's consumption by c_j. However, not all the goods are consumed—some goods are used for conducting visits and matching with a seller. This means that the number of goods consumed is less than the amount of goods purchased, which we denote y_j. Thus, there is a wedge between goods consumed and goods purchased. We need to assess how large this matching wedge is, and in doing so how consumption and purchases are linked.

Suppose buyer j conducts v_j visits with the aim of consuming c_j goods. Then the total number of goods purchased is $y_j = c_j + \kappa v_j$, as each visit requires κ goods. Where do the purchases come from? Each visit leads to a purchase only with probability $q(\theta)$. Thus, to obtain y_j purchases, a buyer needs to make $v_j = y_j/q(\theta)$ visits. Here we assume that there is no uncertainty at the buyer level: a fraction $q(\theta)$ of the visits provides a good. Hence, output and consumption are related by

$$y_j = c_j + \frac{\kappa y_j}{q(\theta)}. \tag{4.6}$$

Solving for y_j in the equation, we find that it is proportional to c_j:

$$y_j = [1 + \tau(\theta)]\, c_j, \tag{4.7}$$

where the matching wedge $\tau(\theta)$ is given by

$$\tau(\theta) = \frac{\kappa}{q(\theta) - \kappa}. \tag{4.8}$$

The matching wedge (4.8) is the gap between the amount of goods consumed and the amount of goods purchased. The gap is caused by the matching process: it arises due to the matching cost associated with each visit. Overall, to consume one good, a buyer must purchase $1 + \tau(\theta)$ goods. Reciprocally, if a buyer purchases one good, they only consume $1/[1 + \tau(\theta)]$ goods.

4.6.2. Properties of the matching wedge

We now discuss the properties of the matching wedge $\tau(\theta)$.

First, we restrict the analysis to $\theta \in (0, \bar{\theta})$ to ensure that the matching wedge is positive and finite, where the upper bound $\bar{\theta}$ is defined by

$$\bar{\theta} = q^{-1}(\kappa). \tag{4.9}$$

Since q is strictly decreasing from $(0, \infty)$ to $(0, q(0))$, the inverse q^{-1} is well defined on $(0, q(0))$. Hence $\bar{\theta}$ exists as long as $\kappa < q(0)$, which we have assumed earlier.[2] Moreover, as q is strictly decreasing, $q(\theta) > q(\bar{\theta}) = \kappa$ for any $\theta < \bar{\theta}$, and we verify that $\tau(\theta)$ is positive and finite for $\theta \in (0, \bar{\theta})$.

Next, we can see that the matching wedge is increasing in tightness, since it is the composite of two decreasing functions: $q(\theta)$ is decreasing in θ and $\kappa/(x - \kappa)$ is decreasing in x.

Moreover, $\tau(\theta) \to +\infty$ when $\theta \to \bar{\theta}$. The divergence $\tau(\theta) \to \infty$ reflects the fixed-point logic that appears in (4.6): matching resources must themselves be purchased through additional matching. As tightness approaches $\bar{\theta}$, the market is so tight that all goods purchased are devoted to matching: no goods are left for consumption.

We plot the matching wedge against tightness to visualize how it behaves (figure 4.1E). With the calibration used in the plot, the vertical asymptote of the wedge occurs at $\bar{\theta} = 4.9$.

4.6.3. Advantages of assessing the matching cost in terms of goods

We assume that the matching cost incurred by buyers is assessed in terms of goods. This means that to pay for the visits, buyers have to purchase an extra amount of goods that covers the visit costs. We make this choice for a few reasons, which we review here.

First, the assumption is tractable because then, we do not need to introduce an extra good to measure the matching cost. We can just focus on one market without thinking

[2] For the urn-ball and CES matching functions, $q(0) = 1$, so $\kappa < 1$ is sufficient.

about another market where that other good would be traded. The fact that the market is self-contained is helpful to think about welfare and efficiency.

Second, the assumption is portable: it allows us to think about any slackish markets, without having to specify the matching cost in different ways in different markets. In particular, it allows us to have isomorphic product and labor markets, which is especially convenient in the business-cycle model of chapter 15.

Third, the assumption is not unrealistic. In the labor market, a huge amount of labor is devoted by firms to recruiting workers—including headhunters hired by firms, human-resource workers within firms, and of course regular workers spending time reading resumes and interviewing future colleagues. In various markets for services, part of what is purchased is the service of a broker or agent who helps source the desired services. For instance, in the temporary staffing industry, the staffing agency is paid in part to recruit workers with the appropriate skills, experience, and availability. In the construction industry, the project manager is paid in part to find the appropriate construction workers required for the various tasks to complete the project (plumber, roofer, painter, landscaper, and so on). In the wedding industry, the wedding planner is paid in part to find the appropriate caterer, florist, band, dressmaker, and so on. It is maybe less clear how the assumption applies to markets for actual goods. However, in that case, we can introduce the matching cost in a different way: as the cost of a sample that the buyer procures to determine whether they want to purchase the good or not. This alternative representation is equivalent to the representation used in the model, and it applies to many goods markets (appendix E).

4.7. Market demand

The cornerstone of the demand side is the buyer's optimization problem: choosing how much to consume so as to maximize utility subject to a budget constraint. To build the market demand, we first need to specify buyers' utility function, then their budget constraint, before solving the maximization problem and deriving individual demands.

4.7.1. Buyer's utility function

We assume that buyers have utility over two things. First, buyer j has utility over consumption of the goods sold on the market, $c_j \geq 0$. Because we want to have a demand for goods, buyers must have the choice between spending on goods and spending on something else. If there is no choice, the concept of demand is not well-defined. We introduce money as the alternative that makes demand well-defined in our static environment. We assume that buyers derive utility from money balances, denoted b_j.

To further simplify the analysis, we assume that the buyer's utility function is quasi-linear. Specifically, we assume that buyer j has utility over consumption and money balances:

$$\mathcal{U}(c_j, b_j) = ac_j^{1-\alpha} + b_j, \tag{4.10}$$

where $a > 0$ governs the taste for goods relative to money and $\alpha \in (0, 1)$ governs the concavity of the utility over goods, which determines diminishing marginal utility of consumption.

4.7.2. Buyer's budget constraint

Buyer j maximizes their utility, but of course subject to a budget constraint. We assume that to start with, each buyer has an endowment of money B_j, which they use to hold money or buy goods. The budget simply says buyers can spend the endowment on goods or keep some of the endowment in money. In order to consume c_j goods, buyer j must purchase $[1 + \tau(\theta)]c_j$ goods at a price p. Hence, the expenditure to purchase goods is $p[1 + \tau(\theta)]c_j$. The buyer's budget constraint therefore is:

$$b_j + p\left[1 + \tau(\theta)\right] c_j = B_j. \tag{4.11}$$

The matching structure introduces a new term into the buyer's budget constraint: the matching wedge, $\tau(\theta)$. It appears because visits to sellers are costly. The wedge is specific to the slackish world; in the Walrasian world, anyone can buy any quantity at the given market price without any other cost, so there would be no matching wedge.

This new element modifies the buyer's behavior. Nevertheless, we can still treat the problem as we usually do: maximizing utility subject to a budget constraint. It's just that, in this case, the budget constraint is slightly modified.

4.7.3. Buyer's problem

We now need to solve the buyer's problem. This is a crucial element of the model because buyers are making the only behavioral decision here: how many goods to consume. Equivalently, buyers decide how many sellers to visit: once they have determined the number of visits, given the probability that a visit is successful, they also determine their consumption.

Buyer j's problem is to maximize their utility (4.10) given the budget constraint (4.11), taking aggregate variables as given: market price, p, and market tightness, θ. The easiest way to solve the problem is to express the buyer's money balances as a function of consumption, by reworking the budget constraint:

$$b_j = B_j - p\left[1 + \tau(\theta)\right] c_j.$$

Then, substituting these money balances into the buyer's utility function, the buyer's problem becomes $\max_{c_j \geq 0} \mathcal{U}(c_j, B_j - p[1+\tau(\theta)]c_j)$, or

$$\max_{c_j \geq 0} a c_j^{1-\alpha} + B_j - p\left[1 + \tau(\theta)\right] c_j. \tag{4.12}$$

Note that we do not require money balances to be positive. Buyers may therefore spend more than their endowment and end up with negative balances—which we would interpret as borrowing instead of saving. We do this to avoid the complications that would arise if some buyers with low endowment were up against their borrowing constraint while other buyers with high endowment were not.

Here we assume that sellers and buyers take market tightness as given. This is quite a natural assumption because market tightness is the ratio of aggregate number of visits to aggregate number of goods for sale. As each market participant is small relative to the size of the market, it cannot influence tightness. We also assume that sellers and buyers take the market price as given; we will see when we discuss the formation of the market price in section 4.8 that this assumption is also reasonable.

Before we solve this maximization problem, let us briefly verify that it is well behaved. Since $\alpha > 0$, the first term is concave in c_j. The second term is linear in c_j, so concave in c_j. The objective function is the sum of two concave functions, so it is concave in c_j. In addition, the objective function's domain, $[0, \infty)$, is convex. Accordingly, we are solving an unconstrained concave maximization problem—which we can conveniently do via first-order condition.

Indeed, with this type of problem, the first-order condition is sufficient to find the global maximum. We therefore set the derivative of the objective function to 0 to determine the optimal c_j. The first-order condition directly yields

$$(1-\alpha) a c_j^{-\alpha} = p\left[1 + \tau(\theta)\right]. \tag{4.13}$$

This condition says that at the optimum the buyer is indifferent between consuming one additional unit of good (which provides utility $(1-\alpha)ac^{-\alpha}$) and holding an additional $p[1+\tau(\theta)]$ units of money (which provides utility $p[1+\tau(\theta)]$), which is the additional amount of money available to the buyer if they forego consumption.

Therefore, at the optimum, the buyer cannot improve their utility by consuming slightly more or slightly less: they are doing the best they can. This optimality condition is standard: it equalizes the marginal rate of substitution between consumption and money balances, $(\partial \mathcal{U}/\partial c)/(\partial \mathcal{U}/\partial b)$, with the price of consumption in terms of money. But, since money is the numeraire, and the marginal utility of money balances is 1, the condition simply equalizes the marginal utility from consumption, $\partial \mathcal{U}/\partial c$, with the price of consumption, $p[1+\tau(\theta)]$.

We now rewrite the first-order condition to get an expression for the buyer's desired consumption of goods as a function of the market price and tightness:

$$c_j = \left[\frac{(1-\alpha)a}{p} \cdot \frac{1}{1+\tau(\theta)}\right]^{1/\alpha}.$$

The consumption has intuitive properties. If the market price is higher, so goods are more expensive, the buyer consumes fewer goods. If the market tightness is higher, so goods are more difficult to find, the buyer consumes fewer goods. And finally if the buyer derives lower utility from goods (lower a), the buyer consumes fewer goods.

From the amount of goods consumed, c_j, it is easy to back out the rest of the buyer's behaviors. To consume c_j the buyer must purchase $y_j = [1+\tau(\theta)]c_j$ goods, and must visit $v_j = c_j/[q(\theta)-\kappa]$ sellers. The buyer spends $p[1+\tau(\theta)]c_j$ in total on goods and therefore holds $B_j - p[1+\tau(\theta)]c_j$ in money.

4.7.4. Expression of the market demand

We now aggregate the individual demands for consumption of goods to obtain the market demand for consumption: $c = \int_0^1 c_j \, dj$. We assumed that different buyers have different initial wealth. One might expect this to lead to a complex distribution of consumption choices. However, a crucial property of the quasilinear utility function (4.10) is that it eliminates wealth effects for the choice of c_j. Hence, all buyers choose the exact same amount c_j, regardless of whether they are rich or poor. This means we can aggregate easily. Since there is a mass 1 of buyers, the market demand for consumption $c_d(\theta, p)$ equals the individual demand that we have just derived:

$$c^d(\theta, p) = \left[\frac{(1-\alpha)a}{p} \cdot \frac{1}{1+\tau(\theta)}\right]^{1/\alpha}. \tag{4.14}$$

This equation gives the notional market demand. It indicates how much buyers would like to consume for a given market price and tightness.

However, in our model, we use an effective market demand, which is the amount of goods that buyers actually desire to buy for a given market price and tightness, which is more than the notional demand. The gap arises because the amount of goods that buyers are able to consume is less than the amount of goods that they buy, since, in a slackish market, some of the purchased goods are devoted to matching instead of consumption.

Conveniently, the concepts of market demand and supply are consistent—the supply and demand both measure goods that are traded. Hence, when we solve the model, we can use the condition that supply equals demand, as any good that is sold by a seller is a good that is bought by a buyer. By focusing on the number of trades, we have equality of supply and demand, even though the model is not Walrasian.

Formally, the market demand, $y^d(\theta, p)$, is the total amount of goods that all buyers purchase so as to maximize their utility given the market price, p and the market tightness, θ. To compute this market demand, we translate the amount of consumption desired, given by (4.14), into the amount of purchases desired, using the result that consuming one good requires purchasing $1 + \tau(\theta)$ goods. This means that

$$y^d(\theta, p) = [1 + \tau(\theta)]\, c^d(\theta, p),$$

which then gives the following expression:

$$y^d(\theta, p) = \left[\frac{(1-\alpha)a}{p}\right]^{1/\alpha} \cdot \frac{1}{[1+\tau(\theta)]^{1/\alpha - 1}}. \tag{4.15}$$

The market demand $y^d(\theta, p)$ is the number of goods demanded by buyers given market price and tightness.

4.7.5. Properties of the market demand

Let us turn to the properties of the market demand, given by (4.15). First, we see that it is decreasing in the market price. When the price level is higher, purchasing the good becomes less attractive compared to holding money, so demand is lower.

Similarly, the market demand is decreasing in market tightness. This property is visible in (4.15) because the matching wedge $\tau(\theta)$ is increasing in θ, while $1/\alpha > 1$. Intuitively, when the market is tighter, it is more difficult to buy goods: each visit is less likely to be successful, so buyers must visit more sellers per purchase, which pushes the matching wedge up. Hence, buyers have to devote a larger share of their purchase to matching instead of consumption, which makes consumption less appealing, and reduces demand.

Recall that $\bar{\theta}$ is the value of tightness at which $\tau(\bar{\theta}) = \infty$. We see that at this value, demand is 0. When tightness is 0, on the other hand, demand is given by

$$y^d(0, p) = \left[\frac{(1-\alpha)a}{p}\right]^{1/\alpha} [1+\tau(0)]^{1-1/\alpha}.$$

To wrap up our analysis of the market demand, we plot the demand curve in a simple graph with quantity on the x-axis and tightness on the y-axis (figure 4.1F). The graph depicts the properties of the market demand that we have just derived. This representation of quantity against tightness is specific to slackish markets. It differs from the Walrasian representation of quantity against price. In slackish markets, we plot quantity against tightness because tightness is the core variable that equilibrates the market.

4.8. Market price

Something that we haven't discussed yet is how the price at which goods are traded is determined. We address this important element now.

We are dealing with a slackish market where trades occur in a decentralized fashion, as described by the matching function. In a Walrasian market goods are sold at auction and the price is set by the auctioneer so as to equalize supply and demand. Here all trade occurs in individual buyer-seller pairs, who are free to negotiate the price as they wish. We cannot assume that a centralized auctioneer sets the price, as we do in a Walrasian market. We need to find a way to model how prices are formed in each of the buyer-seller pairs once they have met on the market.

We introduce a price norm that describes how prices are set in all these buyer-seller pairs. The price norm is very general and can take many forms. As we discuss in chapter 6, we may postulate a price that is a rigid function of the model parameters, or we may assume bargaining between buyers and sellers.

For the time being, we simply assume that prices in all trades are identical and are given by a generic price norm $p = p^n(\text{market variables})$. Since all market variables can be expressed as a function of the market tightness, we simplify the expression of the price norm to

$$p = p^n(\theta). \tag{4.16}$$

The goal should be to specify a price norm that reflects how prices are set in practice in the market studied—we want a price norm that is empirically valid. In chapter 5, we review evidence on how prices and wages are set, and in chapter 6, we specify a price norm that encodes these real-world properties.

Instead of assuming an auctioneer, as in Walrasian theory, our decentralized model gives us the freedom to make assumptions that are as realistic as possible about how prices are set in the real world. This is a vast improvement over the Walrasian model, because the assumption of perfectly competitive prices is probably the least realistic and therefore the most problematic assumption used in modern market models. Furthermore, the assumption of perfectly competitive prices has a raft of implications about welfare and policy, which are all erroneous if the pricing assumption itself is erroneous.

As in Walrasian theory we assume that sellers and buyers take prices as given. One might think that this is problematic because a buyer and a seller could bargain the transaction price once they have matched. However, the actual transaction price has no influence on anyone's decisions because the decisions are made before the match is realized. What matters is the price at which buyers and sellers expect to trade. This is what determines their decisions. The bargained price is only a transfer from buyer to seller: it has no effect on traded quantities, since bargaining would never lead to the destruction of a match. For instance if a price is a little lower than expected, the seller takes home a little less income, which means that they hold a little less money, while the buyer spends a little less, which means that they hold a little more money. But the market quantities—visits, sales, consumption, tightness, slack—would be unchanged.[3]

4.9. Solution of the model

Finally, we present a strategy to solve the model. The main step in solving the model is to determine the market tightness, which we do using a supply-equals-demand condition.

4.9.1. Structure of the solution

Now that we have introduced all the elements of the model and made all the assumptions we need, we are finally in a position to actually solve the model. Before we do so, let us first discuss the structure of the solution: the elements that we are looking for and what we can use to find them.

Formally, solving a model means figuring out the values of all the variables in the model as a function of the parameters of the model. Here, this is actually pretty simple. In our slackish market model, we have to find 6 aggregate variables using 6 independent equations, as shown in table 4.1. There are also individual variables in the model, such as consumption c_j, expenditure y_j, money balances b_j, and visits v_j by any individual buyer j, but these can easily be determined from the aggregate variables.

Now that we have the structure of our solution, the challenge is to find a simple way to compute all the variables in the model.

Before we do that, it is useful to realize that the description of the model in table 4.1 is not unique: it is one of many equivalent representations. Indeed, it is possible to reshuffle the independent equations into other equations, which offer alternative ways to solve the model. Different representations might be useful in different situations, depending on the question and property of interest.

The 6 equations form one valid representation of the model. We can algebraically combine and rearrange them to form a different but equivalent set of 6 independent

[3] We could generalize the model by allowing the transaction price to be the price norm plus some white noise. This would not change anything, so to keep things simple, we assume that actual prices are just the price norm, which buyers and sellers take as given.

TABLE 4.1. Structure of the slackish market model

Variable	Name	Condition	Interpretation	Reference
c	Consumption of goods	$c = c^d(\theta, p)$	Utility maximization	(4.14)
p	Price of goods	$p = p^n(\theta)$	Price norm	(4.16)
y	Sales of goods	$y = [1 + \tau(\theta)]c$	Matching wedge	(4.7)
v	Visits	$q(\theta)v = y$	Matching function	(4.1)
θ	Market tightness	$\theta = v/k$	Definition of tightness	(4.2)
u	Rate of slack	$u = 1 - f(\theta)$	Definition of slack	(4.4)

equations. The key is that any valid system yields the same solution. For instance, if we divide $y = q(\theta)v$ by $\theta = v/k$, we get $y/\theta = q(\theta)k$, or $y = \theta q(\theta)k$, which is just $y = f(\theta)k = y^s(\theta)$. This manipulation is telling us that we could determine sales from the market supply. However, then we would need to replace either the condition for v or θ, since otherwise the 3 equations for y, v, and θ would not be independent. Conceptually, if we did not replace the equations for v or θ, we would lose the information about the matching wedge and matching cost in the model solution—the information that used to be conveyed by the condition that $y = [1 + \tau(\theta)]c$.[4]

4.9.2. Strategy to solve the model

To solve the model, we need to figure out the values of the 6 variables that are determined by 6 independent equations (table 4.1). We follow a sequential strategy for solving the model: we first determine the market tightness, θ; then, we compute the values of the 5 remaining variables. We will compute the market tightness in the next section. Here, we describe how the 5 other variables can easily be determined once we know tightness.

First, we know via the price norm that $p = p^n(\theta)$ so we can easily compute p given θ. Similarly, the rate of slack is directly determined by tightness: $u = 1 - f(\theta)$.

For consumption, we simply use the notional market demand: $c = c^d(\theta, p)$. Since the price p is itself a function of tightness, we can determine c if we know θ.

Then, to compute sales, we simply use that $y = [1 + \tau(\theta)]c$. Since c can be determined from θ, we can compute y given θ.

Last, to calculate the number of visits, we have that $v = y/q(\theta)$. Again, since y is a function of tightness θ, we can compute v as soon as we have θ.

[4]For instance, we could determine the number of visits by $y = c + \kappa v$, or $v = (y - c)/\kappa$. That condition would capture properly how the matching cost enters into the model. And this condition contains the same information as the previous equations $y = [1 + \tau(\theta)]c$ and $q(\theta)v = y$, so no information would be lost by replacing the equations for y and v in table 4.1 by $y = y^s(\theta)$ and $v = (y - c)/\kappa$. Indeed, $y = [1 + \tau(\theta)]c$ implies $y = q(\theta)c/[q(\theta) - \kappa]$, which gives $y[q(\theta) - \kappa] = q(\theta)c$, then $q(\theta)(y - c) = \kappa y$, and finally $(y - c)/\kappa = y/q(\theta)$. But then $q(\theta)v = y$ tells us that $v = y/q(\theta)$, so combining both equations we learn that $v = (y - c)/\kappa$, which is just the new equation used to determine v.

We have shown that we can easily determine the values of the remaining 5 variables once we know market tightness. Tightness is the most critical variable and the next step is to figure out how to compute it.

4.9.3. Computing market tightness from demand and supply

We are now ready to compute the market tightness, θ. From it we can determine the values of all the other variables in the model.

First, we need to think about the key decisions in our model. The first is the buyers' purchasing decision given market tightness, which is characterized by the market demand:

$$y = y^d(\theta, p^n(\theta)). \tag{4.17}$$

The market demand gives the amount of output demanded by buyers.

The second important decision is buyers' shopping decision, where the number of visits is chosen such that buyers purchase the desired amount of goods given market tightness:

$$y = vq(\theta).$$

By definition of tightness, we know that

$$v = \theta k.$$

So, if we know the number of goods that are supplied to the market, k, and tightness, θ, we can calculate the aggregate number of visits. Then,

$$y = \theta q(\theta)k.$$

Recall that $\theta q(\theta) = f(\theta)$ and $f(\theta)k = y^s(\theta)$, so

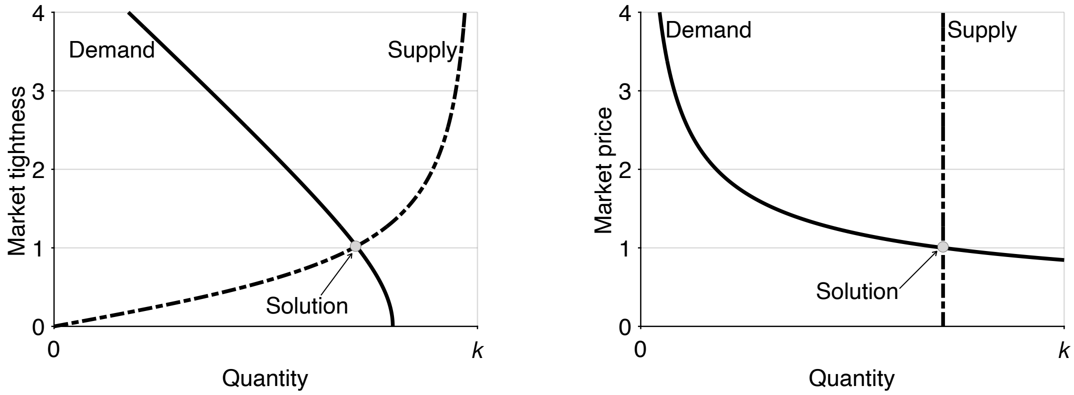
$$y = y^s(\theta), \tag{4.18}$$

which just says that output is given by the market supply too.

Thus, in our model, output is given by both market demand and supply (equations (4.17) and (4.18)). The solution of the model must therefore ensure that market demand equals market supply—otherwise how could output be equal to both market demand and market supply. Accordingly, market tightness must equalize market demand and supply:

$$y^d(\theta, p^n(\theta)) = y^s(\theta). \tag{4.19}$$

The value for market tightness that solves the model can be found at the intersection



A. Solution in tightness-quantity plane

B. Solution in price-quantity plane

FIGURE 4.2. Numerical solution of the slackish market model

The market supply is given by (4.5). The market demand is given by (4.15). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, and $a = 2$. In panel A, the price is set to $p = 1$ and tightness varies. In panel B, tightness is set to $\theta = 1.02$, which is obtained from the intersection of the supply and demand curves in panel A, and p varies.

of the market supply and demand curves. The market supply curve captures the number of trades, for any tightness, given the market capacity and the matching process. The market demand curve captures the number of trades, for any tightness, that maximize the utility of buyers given their budget constraints. Thus, at the solution, the matching process is respected, as well as the utility maximization process.

The supply-equals-demand condition (4.19) is the equivalent of the market-clearing condition of Walrasian theory. The Walrasian market-clearing condition imposes that at the market price, the quantity that buyers desire to buy equals the quantity that sellers desire to sell. This condition is required to ensure the consistency of the Walrasian model because sellers and buyers make their decisions expecting to be able buy and sell any quantity at the quoted price. Here, the supply-equals-demand condition imposes that at the market tightness, the quantity that buyers desire to buy equals the quantity traded through the matching process. The condition ensures the consistency of the slackish model: that buyers visit the appropriate number of shops given how much they desire to consume.

Just as a Walrasian market, a slackish market can be conveniently analyzed via supply and demand curves in a market diagram. Figure 4.2A plots the market supply and demand curves, assuming a simple price norm ($p = 1$). The key difference with a Walrasian market is that it is the market tightness, not the market price, that equilibrates supply and demand. That's why the supply and demand curves are plotted in a tightness-quantity plane, not in the Walrasian price-quantity plane. The tightness that solves the slackish model is found at the intersection of the two curves, just like the price that solves the Walrasian model is found at the intersection of the Walrasian supply and demand curves. In this numerical example, the solution is $\theta = 1.02$.

Of course we can also represent the supply-equals-demand condition in a Walrasian, price-quantity plane (figure 4.2B). In this graph we set the tightness to its solution value: $\theta = 1.02$. The supply curve is just vertical in this plane, because it does not depend on the price. The demand curve is downward sloping. The intersection between supply and demand curves occurs at the price norm, $p = 1$, because we set tightness to the value such that $y^s(\theta) = y^d(\theta, 1)$. Thus, when $p = 1$, supply equals demand. But this graph does not bring new information. It is just repeating what we saw in the tightness-quantity graph: that at $\theta = 1.02$ and $p = 1$, supply equals demand. Furthermore, unlike tightness, the price is not determined by market forces: it is given by the price norm. So the price-quantity graph is not meaningful in a slackish market.

4.9.4. Brief discussion of the solution

To conclude, let us discuss a few things about the solution of the model.

First, just like there are alternative ways to describe the structure of the model, as we saw in table 4.1, there are alternative ways to get the market tightness. An equivalent approach, followed by Michaillat and Saez (2015), is to formulate the market supply and demand in terms of consumption instead of output. Then the market demand $c^d(\theta, p)$, given by (4.14), indicates the number of goods that buyers want to consume. The market supply $c^s(\theta)$ indicates the number of goods traded given the matching process and the amount placed for sale by sellers. This alternative market supply is given by $c^s(\theta) = y^s(\theta)/[1 + \tau(\theta)]$. Condition (4.19) is equivalent to $y^d(\theta, p)/[1 + \tau(\theta)] = y^s(\theta)/[1 + \tau(\theta)]$ and so to $c^d(\theta, p) = c^s(\theta)$. The consumption-based solution makes it easier to think about welfare, but is otherwise slightly less natural, so we follow the output-based approach in this book.

Second, in many models with matching functions, the key decision by buyers is the effort that they put into searching for goods. For instance, in a DMP model, this effort is the vacancies posted by firms. In these search-and-matching models, the solution of the model is organized around the search effort. As we show in appendix F, it is possible to reframe the solution of the model in terms of the number of visits to sellers—which is search effort in this model. The two solutions are equivalent, but the visit-based approach is less convenient analytically.

Third, a natural question is about the realism of the solution concept. When we solve the model, we assume that buyers maximize their utility given market tightness. But how can buyers determine what market tightness is on the market? This is a typical challenge in markets in which buyers and sellers must figure out the values of market variables on which they base their decisions. It is quite a challenging task for buyers to consider the entire market around them and assess what the market tightness will be once all buyers and sellers enter the market.

We discuss these issues further in appendix G and propose a simple resolution. We assume that a statistical agency announces tightness in advance, and that buyers act based on the tightness that has been announced. Then, if the agency announces the tightness equalizing market supply and market demand, the prevailing tightness is the announced tightness. So the agency is correct, and the supply-equals-demand tightness prevails, just like in this chapter.

In fact, even if the statistical agency is unable to build market demand and supply, or unable to equalize both, the agency may find the tightness solving the model via a tatonnement process. Under certain conditions, the model behaves well enough that the tatonnement process converges to the model solution, at which point the tightness quoted by the agency is the same as the tightness realized on the market.

We have now assembled the basic machinery of a slackish market. The core equation of the model is the supply-equals-demand condition, given by (4.19), which determines market tightness. For the time being, we cannot solve the equation explicitly or perform comparative statics because the solution depends on the price norm, $p^n(\theta)$. In the next chapter, we will look at different assumptions for the price norm and solve what the market tightness is in each case and determine how the model responds to different shocks for each norm.

4.10. Model with fixed prices

To complete this chapter, we analyze a slackish market model in which the price norm gives a completely fixed price. The fixed-price model is not only simple but also quite useful because any price that's not fully flexible produces the same qualitative results as a fixed price—with of course different quantitative results—as we will show in chapter 6. Furthermore, the model with a fixed price is the most striking example of a slackish model: all adjustments to shocks occur through market tightness, and none through prices.

4.10.1. Solution of the model

We start by solving the slackish market model when prices are fixed. We assume that the price norm imposes a fixed price $p > 0$ for all goods.

To solve the model, we need to find the tightness θ that equalizes market demand and supply:

$$y^d(\theta, p) = y^s(\theta). \tag{4.20}$$

We first establish the existence and uniqueness of the solution by introducing the excess demand function:

$$z(\theta, p) = y^d(\theta, p) - y^s(\theta). \tag{4.21}$$

With the excess demand function, the solution of the model simply is

$$z(\theta, p) = 0.$$

We know that $y^d(\theta, p)$ is strictly decreasing from $y^d(0, p) > 0$ when $\theta = 0$ to 0 when $\theta = \bar{\theta}$. At the same time, $y^s(\theta)$ is strictly increasing in tightness, starting from 0 when $\theta = 0$. Moreover, both y^d and y^s are continuous in tightness.

Thus, we know that the excess demand function z is continuous in θ, and it declines from $z(0, p) = y^d(0, p) > 0$ to $z(\bar{\theta}, p) = -y^s(\bar{\theta}) < 0$ when θ increases from 0 to $\bar{\theta}$. Thus, by Bolzano's theorem, the equation $z(\theta, p) = 0$ has a solution on $(0, \bar{\theta})$ (result A.1). Moreover, since the excess demand function is strictly decreasing, we know that the solution is unique.

This argument is illustrated in figure 4.3B.

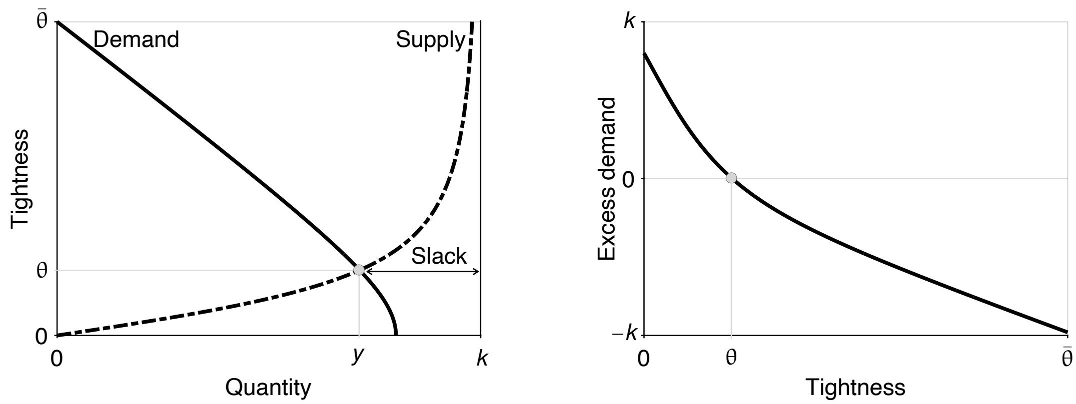
4.10.2. Graphical representation of the solution

To understand the mechanics of the model, and later derive comparative statics, it is very helpful to represent the solution of the model in a tightness-quantity plane. We know that the market demand $y^d(\theta, p)$ is strictly decreasing from $y^d(0, p) > 0$ when $\theta = 0$ to 0 when $\theta = \bar{\theta}$. At the same time, the market supply $y^s(\theta)$ is strictly increasing in tightness, starting from 0 when $\theta = 0$. The market demand curve and market supply curve are illustrated in figure 4.3A. We confirm that the curves $y^d(\theta, p)$ and $y^s(\theta)$ cross once for $\theta \in (0, \bar{\theta})$. The crossing point is the solution of the model.

Comparing figure 4.3A to figure 4.3B, we see that the supply-demand representation conveys more information than the excess-demand representation, which is why we use it throughout the book. The supply-demand representation provides not only the tightness θ that solves the model, but also the corresponding output y, and the amount of slack in the market, $k - y = u(\theta)k$. In particular, we see that in a slackish market, although supply equals demand, there always is some slack: there always are some goods that are available for sale but not sold.

4.10.3. Comparative statics

Now that we have solved the model, we can look at comparative statics: how the model responds to shocks to various parameters. We use our graphical supply-demand representation to visualize the behavior of the model in response to the shocks.



A. Supply-demand representation B. Excess-demand representation

FIGURE 4.3. Solution of the slackish market model with fixed prices

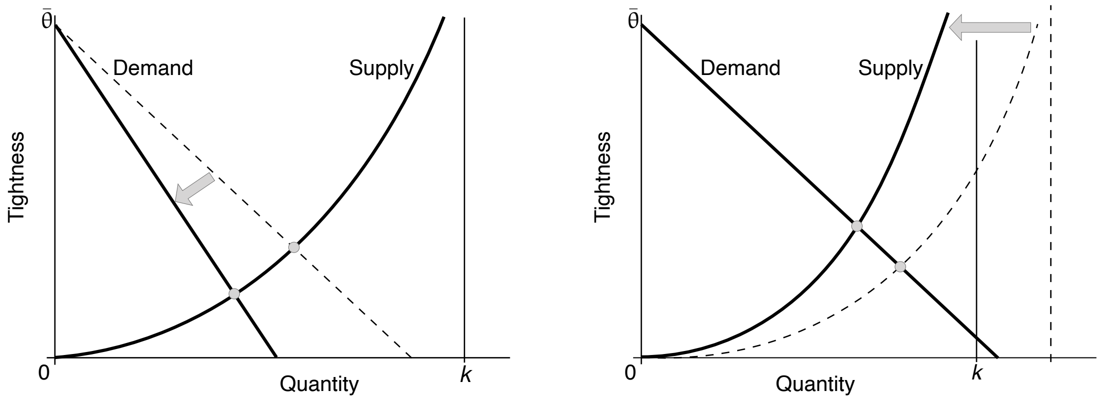
The market supply is given by (4.5). The market demand is given by (4.15). The excess demand is given by (4.21). The matching function is CES, given by (3.12). Parameters are set to $\sigma = 2$, $\kappa = 0.2$, $k = 1$, $\alpha = 0.5$, $a = 2$, and $p = 1$.

Let us start by looking at a negative demand shock: a decrease in preference for goods, a. We start with a high preference, and then reduce the parameter value to a low preference in order to see how the solution of the model changes.

From figure 4.4A, we see that a negative demand shock brings the demand curve inward, which results in a lower value of both tightness θ and sales y. This was to be expected: as buyers value goods less, they aim to consume and therefore purchase fewer goods. They therefore visit fewer sellers, which reduces tightness and the amount of goods sold. The amount of slack in the market increases as it becomes harder for sellers to sell their goods. Note that at this stage we cannot say what happens to consumption. Although purchases decline, the matching wedge $\tau(\theta)$ also decreases, so a larger share of purchases is devoted to consumption. That is, buyers purchase fewer goods but devote a larger fraction of them to consumption. The reason is that with lower demand, visits are more likely to be successful, so fewer visits are required per purchase. Thus, the response of consumption depends on the specific situation, as we clarify in chapter 9.

Another way to be convinced that tightness must fall after a negative demand shock is to assume that tightness remains the same and realize that we reach a contradiction. Since tightness and price remain the same, but the parameter a is lower, we know that buyers want to purchase fewer goods. However, through the matching process, if tightness is the same, the number of goods sold remains the same. So here's the contradiction: if tightness remains the same, it is not possible that both buyers maximize utility and the matching process is satisfied. In other words, keeping the same tightness as before is not a solution of the model.

Next we turn to a negative supply shock: a decrease in the market capacity, k. We start



A. Negative demand shock B. Negative supply shock

FIGURE 4.4. Comparative statics in the slackish market model with fixed prices

The market supply is given by (4.5). The market demand is given by (4.15). The price is fixed to p. A negative demand shock is a reduction in the preference for goods a. A negative supply shock is a reduction in market capacity k.

with a high capacity, and then change the parameter value to a lower capacity in order to see how the model solution changes. In figure 4.4B, we see that a decrease in capacity brings the supply curve inward. What happens to the solution of our model? The new value of sales y is lower than it was before, while the new value of tightness θ is higher. This makes sense because if fewer goods are supplied, through the matching process, fewer goods are bound to be purchased. And because there is more competition among buyers, tightness is bound to increase. Hence slack falls in the market, both in the sense that the share of goods that remain unsold, $u(\theta)$ is lower, and in the sense that the number of goods that remain unsold, $u(\theta)k$ is lower. Since tightness is higher, the matching wedge $\tau(\theta)$ also increases, so a larger share of fewer purchases is devoted to matching. In that case, we know that consumption decreases. What we do not know here is what happens to the number of visits. Tightness is higher so visits are less likely to be successful, which means mechanically that buyers would have to visit more sellers to keep the amount of purchases constant, but which also means that purchases become more complicated and thus less desirable. As we see in appendix F, visits may go up or down after a negative supply shock.

To summarize, both negative demand and supply shocks lead to lower sales; the key difference is that negative demand shocks lead to lower tightness while negative supply shocks lead to higher tightness. After a negative demand shock, it becomes harder to sell goods, while after a negative supply shock, the fewer goods that are for sale are sold more easily. The price does not respond at all to the shocks: it is tightness that absorbs all the shocks and re-equilibrates the market.

4.11. Summary

In this chapter we introduce the basic model of a slackish market: a market in which slack is the central mechanism for equalizing supply and demand. Unlike a Walrasian market, where the price adjusts to equilibrate supply and demand, in a slackish market, the market tightness—here the ratio of buyers' visits to goods for sale—is the variable that equilibrates the market.

The model is built around sellers with a fixed capacity of goods and buyers who must visit sellers to make purchases. The number of trades is determined by a matching function, which captures the complex, decentralized process of buyers finding sellers. Due to the properties of this function, the probability of selling and the probability of buying depend solely on the market tightness. A tighter market favors sellers but harms buyers. The rate of slack, which is the share of goods that remain unsold, is a decreasing function of tightness. The matching wedge, which represents the additional goods buyers must purchase to cover the costs of visiting sellers, is an increasing function of tightness. Importantly, the matching wedge creates a gap between what is purchased and what is consumed.

Buyers maximize their utility, which yields a downward-sloping market demand curve when quantities are plotted against tightness. The market supply is upward-sloping in a tightness-quantity plane because more tightness means a higher probability of selling goods.

The model is solved by finding the market tightness that equalizes market demand and supply. This tightness then determines all other market outcomes, including output, consumption, and the rate of slack. The price at which goods are traded is set by a price norm, which can take various forms.

In the simple case with fixed prices, tightness adjustments absorb all shocks. Negative demand shocks reduce tightness and then sales because of increased slack. By contrast, negative supply shocks increase tightness so they reduce sales despite lower slack.

Bibliography

Barro, Robert J. and Herschel I. Grossman. 1971. "A General Disequilibrium Model of Income and Employment." *American Economic Review* 61 (1): 82–93. `http://www.jstor.org/stable/1910543`.

Michaillat, Pascal and Emmanuel Saez. 2015. "Aggregate Demand, Idle Time, and Unemployment." *Quarterly Journal of Economics* 130 (2): 507–569. `https://doi.org/10.1093/qje/qjv006`.